

AIRBUS

CUSTOMIZATION	AIRCRAFT TYPES	DOCTYPES	REVISION DATE	REVISION NUMBER	TITLE
LDA	A318 A319 A321 A320	AMM	01-Feb-2023	9	72-21-00-320-003-A - Blending of the Fan Rotor Blades
TAIL NUMBER - MSN - FSN:					
Print date: 2023-03-15 02:56:14					

**** ON A/C FSN 001-100, 151-400, 751-800**

TASK 72-21-00-320-003-A
Blending of the Fan Rotor Blades

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

1. Reason for the Job
Self explanatory
2. Job Set-up Information



A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	FILE - SMOOTH
No specific	AR	LIGHT SOURCE - WHITE
No specific	AR	MAGNIFYING GLASS POWER 10X
No specific	AR	WARNING NOTICE(S)
856A2682G01	1	EDDY CURRENT INSTRUMENT
856A2949	1	TEMPLATE, CHECKING - FAN BLADE LEADING EDGE

B. Consumable Materials

REFERENCE	DESIGNATION
CP1039	acetone
CP1041	isopropyl alcohol
CP2011	stoddard solvent
CP2189	durite G422
CP2191	durite G421
CP2559	ardrox 552
CP2560	turco ind 79
CP2566	mag-chem teksol
CP2584	degreasol 99 R
CP2656	mag-chem skysol

C. Work Zones and Access Panels

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ZONE/ACCESS	ZONE DESCRIPTION
400	POWER PLANT, NACELLES AND PYLONS

D. Referenced Information

REFERENCE	DESIGNATION
Ref. 70-51-00-230-002-A	Fluorescent-Penetrant Inspection-Portable-Water Washable
Ref. 70-52-00-250-002-A	Eddy-Current Inspection of the Fan Blade Leading Edge
Ref. 72-21-00-000-007-A	Removal of the Fan Rotor Blades
Ref. SPM 704111 (58828)	
Ref. SPM 704112 (58828)	
Ref. Fig. Blend Repair of the Fan Rotor Blade Airfoil	
Ref. Fig. Fan Blade Blending Repair limits	
Ref. Fig. Typical Examples of Unacceptable Leading Edges Contours	

3. Job Set-up

SUBTASK 72-21-00-941-075-A

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1 (2).

4. Procedure

[Ref. Fig. Blend Repair of the Fan Rotor Blade Airfoil](#)

[Ref. Fig. Fan Blade Blending Repair limits](#)

[Ref. Fig. Typical Examples of Unacceptable Leading Edges Contours](#)

SUBTASK 72-21-00-020-069-A

- A. Remove the damaged fan blades, if necessary [Ref. AMM TASK 72-21-00-000-007](#).

SUBTASK 72-21-00-220-054-A

B. Examine the Fan Blades.

- (1) Leading edge and trailing edge in the critical root area.

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- (a) Find the critical root area:
 - 1 The root area longitudinal dimension is 2.36 in. (60 mm).
- (2) Mid span shroud area.
 - (a) Find the critical area for the mid span shroud:
 - 1 The critical areas are at 1.57 in. (40 mm) from the leading edge and at 1.18 in. (30 mm) on each side of the mid span shroud.
 - 2 The critical areas are at 0.63 in. (16 mm) from the bottom of the mid span shroud and at 0.39 in. (10 mm) from the mid span shroud trailing edge.
- (3) Leading edge and trailing edge repair in non critical areas.
 - (a) The leading edge dimensions:
 - 1 Taper the blend repair for area A and area B as follows:
 - a In area A, the blend repair is not more than 0.08 in. (2 mm).
 - b In area B, the blend repair is not more than 0.339 in. (8.6 mm).
 - c When the distance (Dim. O) between two blend repairs is less than or equal to five times the depth of the smallest of these blends, tangentially blend out the bottom of the two blends.
 - d The total length of blend repairs permitted on the leading edge is equal to 2.76 in. (70 mm) per blade.
 - (b) The leading edge (areas A and B) blend repairs are permitted with these limits:
 - 1 Make sure that after you do the blending repair, the edge thickness is not less than the initial thickness.
 - 2 The longitudinal direction of the repair for area A must be equal to or more than ten times the depth (Dim. P) of the damage.
 - 3 The longitudinal direction of the repair for area B must be equal to or more than eight times the depth (Dim. P) of the damage.
 - 4 The minimum radius R4 is equal to or more than fifteen times of depth (Dim. P) for area A.
 - 5 The minimum radius R4 is equal to or more than ten times of depth (Dim. P) for area B.
 - 6 The radius R3 for the blend repair on either side of the initial damage is a minimum radius blend of 0.2 in. (5 mm) and a maximum radius blend of 0.31 in. (8 mm).

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7 If the distance (Dim. O1) between the blend repair and the blade tip is not more than two times the depth (Dim. P) of the repair, do the step:

- a Blend the bottom area of the repair tangentially from the depth of the area to the blade tip and blend the tip to the blending tip radius R1 with a minimum radius blend of 0.02 in. (0.5 mm) and a maximum radius blend of 0.67 in. (17 mm) at R1 minimum possible value.

(c) The trailing edge dimensions:

1 Taper the blend repair as follows:

- a The blend repair is not more than 0.059 in. (1.5 mm).
- b When the distance (Dim. O) between two blend repairs is less than or equal to five times the depth of the smaller repair area, blend the two repairs together to a depth of a line which tangentially connects the bottom areas.
- c The total length of blend repairs permitted on the trailing edge is equal to 1.77 in. (45 mm) per blade.

(d) Trailing edge blend repairs are permitted with these limits:

- 1 Make sure that after you do the blending repair, the edge thickness is not less than the initial thickness.
- 2 The longitudinal direction of the repair must be equal to or more than ten times the depth (Dim. P) of the damage.
- 3 The minimum radius R4 is equal to or more than fifteen times the depth (Dim. P).
- 4 The radius R3 for the blend repair on either side of the initial damage is a minimum radius blend of 0.2 in. (5 mm) and a maximum radius blend of 0.31 in. (8 mm).
- 5 If the distance (Dim. O1) between the blend repair and the blade tip is not more than two times the depth (Dim. P) of the repair, do this step:
 - a Blend the bottom area of the repair tangentially from the depth of the area to the blade tip and blend the tip to the blending tip radius R2 with a minimum radius blend of 0.02 in. (0.5 mm) and a maximum radius blend of 0.43 in. (11 mm) at R2 minimum possible value.

(4) Leading edge and trailing edge repair in critical areas.

(a) The leading edge and trailing edge dimensions:

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- 1 Taper the blend repair as follows:
 - a The blend repair for the leading edge is not more than 0.04 in. (1 mm).
 - b The blend repair for the trailing edge is not more than 0.059 in. (1.5 mm).
- (b) The leading edge and trailing edge blend repairs are permitted with these limits:
 - 1 Taper the blend repair as follows:
 - a The longitudinal direction of the repair must be equal to or more than twenty times the depth (Dim. P) of the damage.
 - b The minimum radius R4 is equal to or more than thirty times the depth (Dim P).
- (5) Blade tip repair (area C).
 - (a) The blend repair does not extend more than 0.04 in. (1 mm).
 - (b) The chordal direction of the repair must be equal to or more than eight times the depth (Dim. P) of the damage.
 - (c) The minimum radius R4 is equal to or more than ten times the depth (Dim. P) of the damage.
 - (d) The radius R3 for the blend repair on either side of the initial damage is a minimum radius blend of 0.2 in. (5 mm) and a maximum radius blend of 0.31 in. (8 mm).
 - (e) The radius R1 blend repair between the leading edge and the tip is not less than 0.02 in. (0.5 mm) and not more than 0.67 in. (17 mm).
 - (f) The radius R2 blend repair between the trailing edge and the tip is not less than 0.02 in. (0.5 mm) and not more than 0.43 in. (11 mm).
 - (g) When the distance (Dim. O) between two blend repairs is less than or equal to five times the depth (Dim. P) of the repair, blend the two repairs together to a line which tangentially connects the bottom areas.
 - (h) If the distance (Dim. O1) between the blend tip repair and the blade leading edge is less than two times the depth (Dim. P) of the repair, do this step:
 - 1 Blend the bottom of the repair tangentially across to the blade leading edge and blend the tip to the blending tip radius R1 minimum possible value.
 - (i) If the distance (Dim. O1) between the blend tip repair and the blade trailing edge is less than two times the depth (Dim. P) of the repair, do this step:

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- 1 Blend the bottom of the repair tangentially across to the blade trailing edge and blend the tip to the blending tip radius R2 minimum possible value.
- (j) The total chordal length of the blend repairs permitted on the blade tip is equal to 0.8 in. (20 mm) per blade.
- (6) Fan blade airfoil blend repairs.
 - (a) The convex or concave blend repair:
 - 1 Repairs are permitted on the surface with this limit:
 - a Do not repair inside the blade critical areas.
 - 2 If the distance (Dim. U) is not less than 0.079 in. (2 mm), do this step:
 - a The convex or concave blend repair depth becomes not more than 0.024 in. (0.6 mm).
 - 3 The concave or convex blend repair minimum radius R5 is not less than five times the depth (Dim. K).
 - 4 The concave or convex blend repair area length is the sum of dimensions (Dim. S) and (Dim. T):
 - a The total length of (Dim. S) and (Dim. T) for a blend repair is not more than 0.425 in. (10.8 mm).
 - b The chordal length (Dim. S) for a blade is at least six times the repair length (Dim. K).
 - c The maximum longitudinal length (Dim. T) is two times the length (Dim. S).
 - d The distance (Dim. O2) between two blend repairs must be more than the blend repair length (Dim. S) + (Dim. T) of the largest blend.
 - e The total length (Dim. S) + (Dim. T) of all blend repairs on the convex side must not be more than 1.3 in. (33 mm).
 - f The total length (Dim. S) + (Dim. T) of all blend repairs on the concave side must not be more than 1.3 in. (33 mm).
- (7) Mid span shroud blend repair.
 - (a) The distance (Dim. D) between two blend repairs must be more than five times the two defects largest blend.
 - (b) The distance (Dim. V) for a blend repair is not more than 0.02 in. (0.5 mm).
 - (c) The longitudinal direction (Dim. W) for a blend repair must be equal to or more than eight times the depth (Dim. V).

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- (d) The minimum radius R6 is equal to or more than ten times the depth (Dim. V).
- (e) The radius R3 for the blend repair on either side of the damage is a minimum radius blend of 0.2 in. (5 mm) and a maximum radius blend of 0.31 in. (8 mm).
- (f) The minimum distance (Dim. M) is equal to or more than six times the depth (Dim. V).
- (g) The maximum distance (Dim. N) is two times the length (Dim. M).
- (h) The total length of (Dim. M) and (Dim. N) for a blend repair is not more than 0.35 in. (9 mm).
- (i) The distance (Dim. X) between two blend repairs, located either on the same or opposite sides, must be more than 0.177 in. (4.5 mm).
- (j) The total length (Dim. W) of all blend repairs must not be more than 0.16 in. (4 mm).

SUBTASK 72-21-00-320-052-A

CAUTION: WHEN YOU USE POWER GRINDING OR BLENDING TOOLS, MAKE SURE THAT THE ENGINE BLADE DOES NOT BECOME TOO HOT. IF THE BLADE BECOMES TOO HOT, LET IT COOL BEFORE YOU CONTINUE THE REPAIR. TOO MUCH HEAT DURING GRINDING OR BLENDING CAN MAKE THE BLADE WEAKER.

C. Do the blending operation as follows:

NOTE: If five or more fan blades have been subject to this blending repair, perform trim balance operation. It is recommended that location of repaired blades be recorded to perform trim balance operation.

- (1) Remove the surface damage from the blade airfoil per Ref. SPM 704111 (58828) using a FILE - SMOOTH or strips of durite G422 (Material Ref. [CP2189](#)) or durite G421 (Material Ref. [CP2191](#)) grade 150 or 180, used dry. The blended area must make a smooth transition into the surrounding areas.

NOTE: To prevent axial scratches, always do the blending in a longitudinal direction, parallel with the length of the blade.

- (2) Complete the blending with strips of durite G422 (Material Ref. [CP2189](#)) grade 400 or 800, used dry. The surface finish (roughness) on blended areas must be as close as practical to the original finish.
- (3) If a large or deep blend is required, a low speed power tool equipped with an appropriate titanium grinding tool may be used to shorten the time required to remove the metal per Ref. SPM 704112 (58828) but you must complete with hand blending as told above.

NOTE: Reworked contour must conform to original contour.

[Ref. Fig. Typical Examples of Unacceptable Leading Edges Contours](#)

SUBTASK 72-21-00-110-054-A

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D. Clean the blade:

CAUTION: TITANIUM PARTS REQUIRE THE USE OF NON-HALOGENATED SOLVENTS.

- (1) Clean blade using isopropyl alcohol (Material Ref. [CP1041](#)) or acetone (Material Ref. [CP1039](#)) or stoddard solvent (Material Ref. [CP2011](#)) or ardrex 552 (Material Ref. [CP2559](#)) or turco ind 79 (Material Ref. [CP2560](#)) or mag-chem teksol (Material Ref. [CP2566](#)) or degreasol 99 R (Material Ref. [CP2584](#)) or mag-chem skysol (Material Ref. [CP2656](#)) .

SUBTASK 72-21-00-230-052-A

E. Do an Inspection of Blended Area:

- (1) Do a fluorescent inspection [Ref. AMM TASK 70-51-00-230-002](#) or an eddy current inspection with EDDY CURRENT INSTRUMENT (856A2682G01) (856A2682G01) [1\) Ref. AMM TASK 70-52-00-250-002](#) of blended area for cracks. No crack is permitted.

NOTE: If you did the blending operation on the leading edge in area B, you can delay the fluorescent-penetrant inspection or the eddy current inspection by 10 flight cycles or 25 flight hours (whichever comes first). In this case, you must do a visual inspection with a MAGNIFYING GLASS POWER 10X under LIGHT SOURCE - WHITE, to make sure sure that the damage is fully removed.

- (a) If necessary, repeat blending and inspection until no crack is detected. Refer paragraph A. to paragraph E. (1).
- (2) If you have a template, TEMPLATE, CHECKING - FAN BLADE LEADING EDGE (856A2949G01) (856A2949G01), use it after the blending of the leading edge as follows:
 - (a) Put it on the leading edge of the blade on concave side so that the tabs touch the leading edge and the mid span shroud fillet radius with airfoil.

5. Close-up

SUBTASK 72-21-00-869-070-A

A. Make sure that the work area is clean and clear of tools and other items.

SUBTASK 72-21-00-942-069-A

B. Close Access

- (1) Remove the access platform(s).
- (2) Remove the WARNING NOTICE(S).

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MATERIAL: 6-475 (TANV PQ)

COATING: - Ep-800 THERMAL SPRAYED COATING AND DRY FILM LUBRICANT ON COMPOSITE MATERIAL SURFACES
- TUNGSTEN CARBIDE ON BRONZE SURFACES

3-508-MP-0417-00-0

N_MM_722100_8_0100_01_00

Figure 72-21-00-991-32900-06-A (SHEET 1) - Blend Repair of the Fan Rotor Blade Airfoil
**** ON A/C FSN 001-100, 151-400, 751-800**

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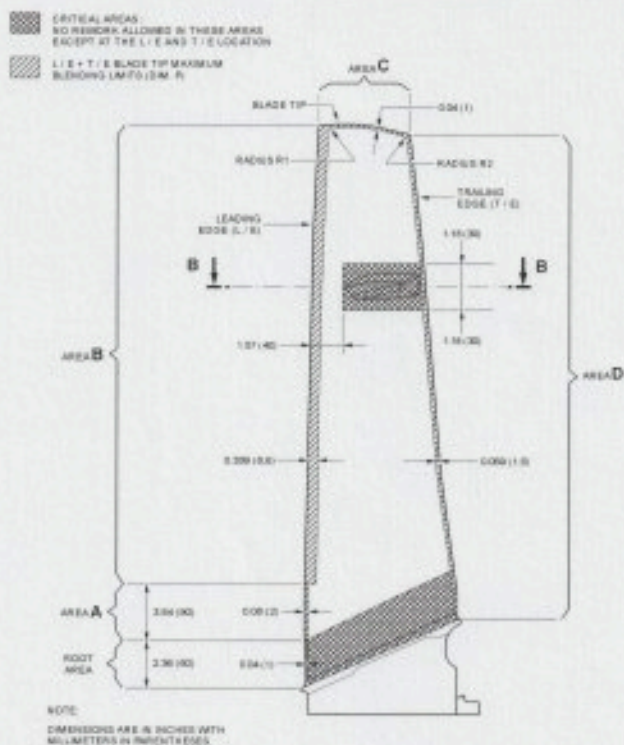

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Figure 72-21-00-991-33300-06-A (SHEET 1/3) - Fan Blade Blending Repair limits
 ** ON A/C FSN 001-100, 151-400, 751-800

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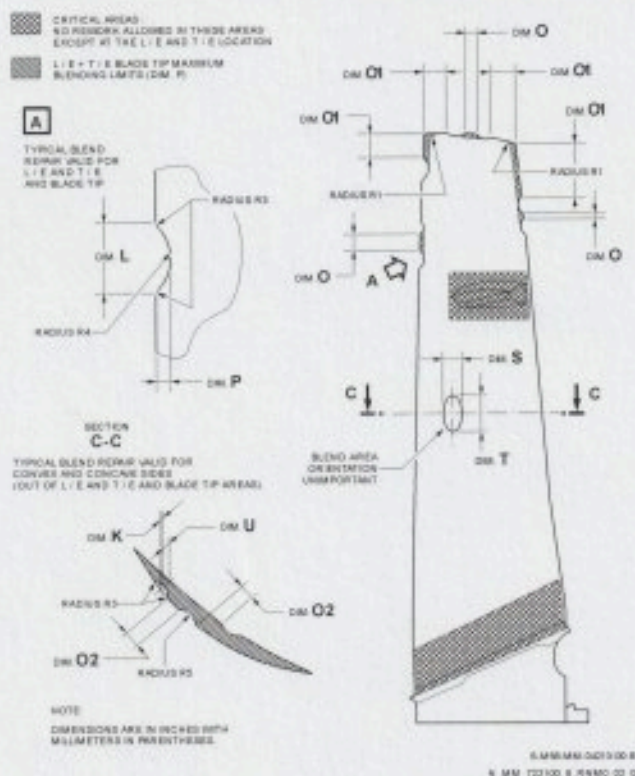


Figure 72-21-00-991-33300-06-A (SHEET 2/3) - Fan Blade Blending Repair limits
 ** ON A/C FSN 001-100, 151-400, 751-800

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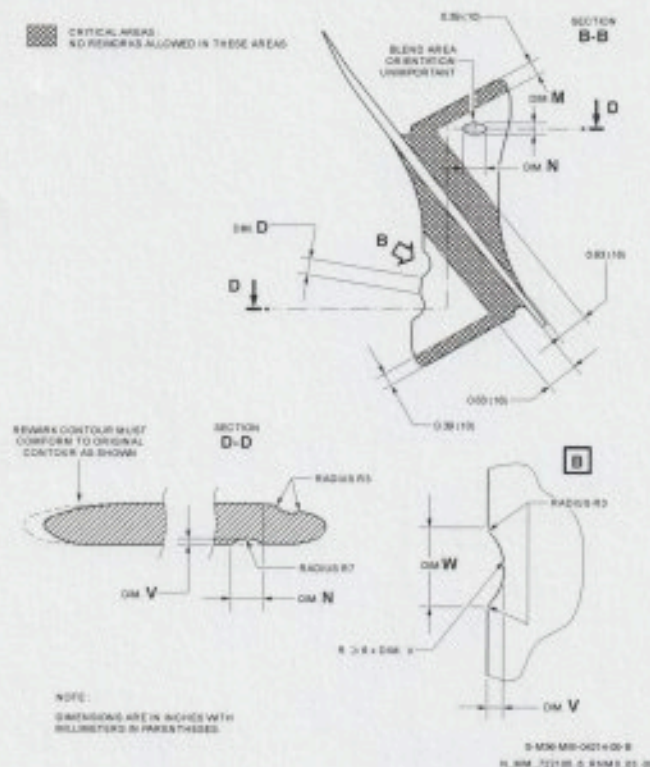


Figure 72-21-00-991-33300-06-A (SHEET 3/3) - Fan Blade Blending Repair limits
**** ON A/C FSN 001-100, 151-400, 751-800**

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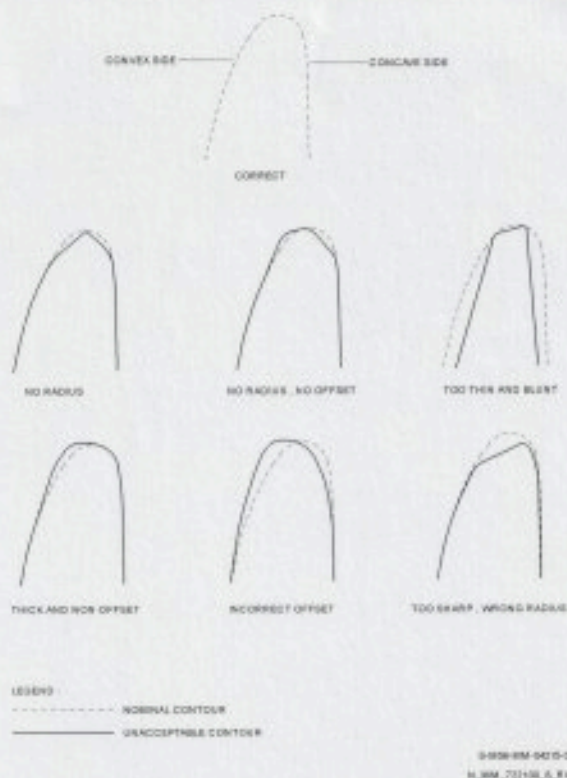


Figure 72-21-00-991-33400-06-A (SHEET 1) - Typical Examples of Unacceptable Leading Edges Contours
**** ON A/C FSN 001-100, 151-400, 751-800**